

Recap.

$$\textcircled{1} P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

"Add odd, minus even".

$$\textcircled{2} F_X(x) = P(X \leq x), \text{ "probability of } X \leq x \text{"}$$

$$\textcircled{3} f_X(x) = F'_X(x) = P(X=x), \text{ "probability of } X=x \text{"}$$

$$\textcircled{4} \text{ Given events } A \text{ and } B, P(B) > 0,$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ "conditional probability of } A \text{ given } B \text{"}$$

$\textcircled{5}$ In the space $(\Omega, \mathcal{A})$, given $B \in \mathcal{A}$ with $P(B) > 0$, $\tilde{P}: \mathcal{A} \rightarrow [0,1]$ defined by $\tilde{P}(A) = P(A|B)$ for $\forall A \in \mathcal{A}$, then probability space $(B, \mathcal{A}|_B, \tilde{P}|_B)$ is trace of $(\Omega, \mathcal{A}, P)$ on B .

$$\textcircled{6} B = \bigcup_{i \in I} B_i \text{ and } B_i \cap B_j = \emptyset \text{ for } i \neq j, P(A \cap B) = \sum_{i \in I} P(A \cap B_i) = \sum_{i \in I} P(A|B_i) P(B_i) = \sum_{i \in I} P(A|B_i) P(B_i).$$

$$\textcircled{7} P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}.$$

Exercises.

Ex 1.

$$\text{(i)} \Omega = \{1, 2, 3, 4\}^k, |\Omega| = 4^k.$$

$$E = \{(i_1, \dots, i_k) \in \Omega \mid i_j \in \{1, 2, 3\} \text{ for all } j\}, |E| = 3^k. P(E) = \frac{|E|}{|\Omega|} = \left(\frac{3}{4}\right)^k.$$

$$\text{(ii)} F^c = \{(i_1, \dots, i_k) \in \Omega \mid i_j \in \{1, 3\} \text{ for all } j\}, |F^c| = 2^k. P(F) = 1 - P(F^c) = 1 - \frac{|F^c|}{|\Omega|} = 1 - \frac{2^k}{4^k} = 1 - \left(\frac{1}{2}\right)^k.$$

$$\text{(iii)} E \cap F = \{(i_1, \dots, i_k) \in \Omega \mid i_j \in \{2, 3\} \text{ for all } j\}, |E \cap F| = 2^k. P(E \cup F) = P(E) + P(F) - P(E \cap F) = \left(\frac{3}{4}\right)^k + 1 - \frac{2^k}{4^k} = \left(\frac{3}{4}\right)^k + 1 - \left(\frac{1}{2}\right)^k.$$

Ex 2. Hint: P is a known probability measure.

(i) To prove: $\textcircled{1} \tilde{P}(\emptyset) = 0$. $\textcircled{2} \forall A \in \mathcal{A}: \tilde{P}(A) \geq 0$.

$$\textcircled{3} \{A_i\} \text{ pairwise disjoint, } \tilde{P}\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i \in \mathbb{N}} \tilde{P}(A_i). \quad \textcircled{4} \tilde{P}(\Omega) = 1.$$

$$\textcircled{1} \tilde{P}(\emptyset) = P(\emptyset|B) = \frac{P(\emptyset \cap B)}{P(B)} = \frac{P(\emptyset)}{P(B)} = 0.$$

$$\textcircled{2} \text{ For } \forall A \in \mathcal{A}, \tilde{P}(A) = P(A|B) = \frac{P(A \cap B)}{P(B)} \geq 0.$$

$$\textcircled{3} \tilde{P}\left(\bigcup_{i \in \mathbb{N}} A_i\right) = P\left(\bigcup_{i \in \mathbb{N}} A_i \mid B\right) = \frac{P\left(\left(\bigcup_{i \in \mathbb{N}} A_i\right) \cap B\right)}{P(B)} = \frac{P\left(\bigcup_{i \in \mathbb{N}} (A_i \cap B)\right)}{P(B)} = \frac{\sum_{i \in \mathbb{N}} P(A_i \cap B)}{P(B)} = \sum_{i \in \mathbb{N}} \frac{P(A_i \cap B)}{P(B)} = \sum_{i \in \mathbb{N}} P(A_i|B) = \sum_{i \in \mathbb{N}} \tilde{P}(A_i).$$

$$\textcircled{4} \tilde{P}(\Omega) = P(\Omega|B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1.$$

$$\text{(ii)} \sum_{i \in \mathbb{N}} P(A_i|B_i) P(B_i) = \sum_{i \in \mathbb{N}} \frac{P(A_i \cap B_i)}{P(B_i)} \cdot P(B_i) = \sum_{i \in \mathbb{N}} P(A_i \cap B_i) = P\left(\bigcup_{i \in \mathbb{N}} (A_i \cap B_i)\right) = P\left(A \cap \left(\bigcup_{i \in \mathbb{N}} B_i\right)\right) = P(A \cap B).$$

(iii) Not discussed, see formal solution.

Ex 3.

(i) Spin the barrel: $P(\text{Survive}) = \frac{2}{3}$.

E, E, E, E, B₁, B₂

Pull the trigger: $P(\text{Survive}) = 1 - P(\text{Die})$

E, E, E, B₁, B₂, E

$$= 1 - P(\text{Next is bullet} | \text{This is empty})$$

E, E, B₁, B₂, E, E

$$= 1 - \frac{P(\text{Next is bullet} \cap \text{This is empty})}{P(\text{This is empty})}$$

E, B₁, B₂, E, E, E

$$= 1 - \frac{\frac{1}{6}}{\frac{4}{6}} = \frac{3}{4} > \frac{2}{3}.$$

...

(ii) Card 1: both face red. Card 2: one side red, one side blue.

$X_{\text{card}} = \{1, 2\}$ for "the picked card". $X_{\text{col}} = \{r, b\}$ for "the shown face".

$$P(X_{\text{card}}=1 | X_{\text{col}}=r) = \frac{P(X_{\text{card}}=1 \cap X_{\text{col}}=r)}{P(X_{\text{col}}=r)} = \frac{\frac{2}{4}}{\frac{3}{4}} = \frac{2}{3}.$$

Ex. 4. Hint: Define a new RV to assist.

(i) Define $U \sim \text{Uniform}[0, 1]$, $X = U^2$. $F_X(x) = P(X \leq x) = P(U^2 \leq x) = P(U \leq \sqrt{x}) = \begin{cases} 0 & \text{if } x < 0 \\ \sqrt{x} & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$

(ii) $f_X(x) = F'_X(x) = \begin{cases} \frac{1}{2\sqrt{x}}, & \text{if } x \in (0, 1) \\ 0, & \text{otherwise} \end{cases}$

(iii) $P(X > \frac{1}{4}) = 1 - P(X \leq \frac{1}{4}) = 1 - F_X(\frac{1}{4}) = 1 - \sqrt{\frac{1}{4}} = \frac{1}{2}.$